Solutions to Review Questions for Lecture 9 and 10

1. There are six equally likely spots and one has the $\$1,000,000$. Hence the chance is $1/6$.

2. The chance in a. is the opposite of not getting the $\$1,000,000$ on either spin, and so is $1 - 5/6 * 5/6 = 11/36$. You would win in b. if you hit the spot with the $\$1,000,000$ or were right before this spot. The chance is $2/6 = 1/3$.

3. In question 1, the chance would be $2/6$ and would not depend on which two spots have the $\$1,000,000$. The chance in 2a. is $1-4/6*4/6 = 20/36$ and does not depend on which two spots have the $\$1,000,000$. The chance in 2b. is $3/6$ if the two spots are adjacent and $4/6$ if the two spots are not adjacent.

4. a) There are 5 tickets with white on the top side out of 6 tickets total. Hence the chance is $5/6$.
   b) There are 5 tickets with white on the top side, and 3 of those have black on the bottom side. Hence the chance is $3/5$.

5. a) This is when it always rains on Sunday if it rains on Saturday, and would be 0.3.
   b) This is when it never rains on Sunday if it rains on Saturday, and would be 0. Note that the probabilities add up to $0.9 < 1$, and so this is possible.

6. a) This is when it always rains on Sunday if it rains on Saturday, and would be 0.4.
   b) The sum of the probabilities is 1.2, and so the overlap has to have probability of at least 0.2.

7. a) The maximum possible value would be 0.2 because *A and B and C* is contained in A. An example achieving this maximum would be from the set $\{1, 2, \cdots, 10\}$ where are numbers are equally likely, $A = \{4, 5\}, B = \{1, 2, 3, 4, 5\}, C = \{4, 5, 6, 7, 8, 9, 10\}$.
   b) The minimum possible value would be 0. An example achieving this mimimum would be the set in part a., but with $A = \{1, 2\}, B = \{1, 2, 3, 4, 5\}, C = \{4, 5, 6, 7, 8, 9, 10\}$.
   c) The maximum possible value would be 1. See the example in b.
   d) The minimum possible value would be $P(C) = 0.7$ since *A or B or C* contains C. This would occur if $A = \{4, 5\}, B = \{6, 7, 8, 9, 10\}, C = \{4, 5, 6, 7, 8, 9, 10\}$.

8. a) The maximum possible value would be 0.8 because *A and B and C* is contained in A. This would happen with the set of integers from 1 to 20 where each number is equally likely, A is the set of numbers from 1 to 16, B is the set of numbers from 1 to 18, and C is the set of numbers from 1 to 19.
   b) The minimum possible probability for *B and C* is $0.9 + 0.95 - 1 = 0.85$, and the minimum possible value for the overlap with A is $0.85 + 0.8 - 1 = 0.65$. An example of this would be where B is the numbers from 1 to 18, C is the numbers from 2 to 20, and A is the number 1 along with the numbers from 6 to 20.
   c) The maximum value would be 1. See part *b*.
   d) The minimum value would be 0.95 since *A or B or C* contains C. Make C the numbers from 2 to 20, B the numbers from 3 to 20, and C the numbers from 5 to 20.

9. a) If we drew two cards from a deck many times without replacement and threw out the times where the second card was black, then roughly $25/21$ of the remaining pairs of cards will have a first card that is red.
   b) If we drew two cards from a deck many times without replacement and threw out the times where the first card was black, then roughly $25/21$ of the remaining pairs of cards will have a second card that is red.

10. a) It never rains on both Saturday and Sunday.
    b) If it rains on Saturday, then it always rains on Sunday.